

KUPC Phase 7 Specification

Swipe Engine

Version 1.0 · July 19, 2026 · Status: Draft for sprint planning · Detailed milestone steps

Document Control

Field	Value
Project	KUPC — Kathmandu University Placement Connect
Phase	7 — Swipe Engine
Depends on	Phase 2 (Auth), Phase 3B (swipes / matches schema & repos), Phase 6 (open jobs feed + Discover UI)
Feeds into	Phase 8+ — Chat / conversations; richer Matching UX; notifications
Repos	kupc-backend + frontend (monorepo)
Audience	Backend engineers, frontend engineers, technical reviewers
References	Backend Spec v2.0; PHASE_3B_DOCUMENTATION.md; PHASE_6_DOCUMENTATION.md; KUPC_Phase6_Specification.pdf; frontend/INTEGRATION.md

1. Phase Goal

Turn Discover swipe gestures into durable product behavior: students record **left** (pass) or **right** (interest) decisions on open jobs; already-swiped jobs leave the deck; companies can see inbound interest and reciprocate to create a **match**. Replace local-only Discover swipe state (`// MOCK: Phase 7`) with live APIs backed by Phase 3B swipes and matches tables.

Phase 6 answered: *How does a company post a job, and how does a student find it?*

Phase 7 answers: *How does a student express interest, and when does that become a match?*

2. Why This Phase Now

- Phase 6 shipped open-job discovery; Discover still advances a local deck without writing swipes — reloads reshuffle the same cards.
- `swipes.repository.ts` and `matches.repository.ts` already exist (Phase 3B) — Phase 7 adds the HTTP module and UI wiring, not new core tables.
- Seed data already creates sample swipes/matches; product paths must match that model (student right-swipe + company reciprocation → match).
- Chat (conversations/messages) stays out of scope until matches are created by real user actions.

3. Scope & Milestones

ID	Side	Topic	Outcome
B1	Backend	Swipes module scaffold	Routes, Zod, DTOs, auth gates; stubs until B2
B2	Backend	Record swipe + feed exclusion	POST swipe; Discover omits swiped job IDs
B3	Backend	Undo (optional window)	Delete/reverse recent swipe within TTL if adopted
B4	Backend	Company interest + match	List right-swipes on own jobs; reciprocate → match
B5	Backend	Matches read APIs	Student/company list own matches (no chat yet)
B6	Backend	Swagger + test matrix	OpenAPI + test : phase7 green
F1	Frontend	swipesApi + matchesApi	Typed clients matching Phase 7 contracts

ID	Side	Topic	Outcome
F2	Frontend	Discover live swipe	Like/Nope → API; deck excludes swiped
F3	Frontend	Company interest inbox	See students who liked a job; reciprocate
F4	Frontend	Matches list (live)	Replace mock Matches with API list
F5	Frontend	Polish + docs	INTEGRATION.md + PHASE_7_DOCUMENTATION.md

3.1 Out of scope

- Realtime chat / message sending UI (Phase 8+)
- Applicant kanban / application workflow
- Push notifications beyond optional match notification insert
- pgvector / semantic ranking of the deck
- Admin moderation of swipes
- Changing Phase 3 schema unless a documented gap is found (prefer repository extensions)

3.2 Suggested order

B1 → B2 → B5 (read matches) in parallel with F1. Then B4 + F2–F3. Optional B3 undo after B2 is stable. Finish B6 + F4–F5.

3.3 Existing assets (do not reinvent)

Asset	Location	Role in Phase 7
swipes table + UNIQUE	Phase 3B migrations	One decision per student/job
matches table	Phase 3B migrations	Mutual interest outcome
swipesRepository	src/database/swipes.repository.ts	create / list / find right-swipe
matchesRepository	src/database/matches.repository.ts	create / list / findByTriple
jobs feed	Phase 6 jobs module	Source deck; exclude swiped IDs
Discover UI	frontend DiscoverPage	Replace local advance with API
Seed swipes/matches	scripts/seed.ts	Demo data model reference

4. API Contract (Backend)

All paths under `/api/v1`. Bearer JWT required. Responses use the existing envelope: `{ success, data, message, error }`. Wire naming remains **snake_case**.

4.1 Student swipes

Method	Path	Auth	Description
POST	<code>/swipes</code>	Student	Body: <code>{ job_id, direction: left right }</code> . Resolves <code>company_id</code> from job; rejects non-open / non-approved. Idempotent conflict → 409.
DELETE	<code>/swipes/:jobId</code>	Student	Optional undo (B3). Only within configured window (e.g. 30s) if enabled; else 409.
GET	<code>/swipes/me</code>	Student	Optional history list (debug / profile); not required for Discover MVP.

4.2 Discover feed (Phase 6 extension)

GET `/jobs` must exclude job IDs already present in swipes for the authenticated student (use `swipesRepository.listJobIdsByStudent + repository filter / listOpenForFeed(excludeJobIds)`). Companies calling the feed are unchanged (no swipe exclusion).

4.3 Company interest & match

Method	Path	Auth	Description
GET	<code>/swipes/inbound</code>	Company + verified	Right-swipes on the company's open/closed jobs (student summary + job title).
POST	<code>/matches</code>	Company + verified	Body: <code>{ job_id, student_id }</code> . Requires existing student right-swipe; creates match if missing (idempotent).
GET	<code>/matches/me</code>	Student or Company	List own matches with nested job + counterparty summary.

4.4 DTOs (illustrative)

- `SwipeDto` — `id, student_id, company_id, job_id, direction, swiped_at`
- `MatchDto` — `id, student_id, company_id, job_id, matched_at, nested job + student or company card`
- `InboundSwipeDto` — `swipe + student public summary + job title/status`

4.5 Error codes

Code	HTTP	When
<code>SWIPE_NOT_FOUND</code>	404	Undo / lookup missing
<code>SWIPE_CONFLICT</code>	409	Duplicate swipe (unique violation)
<code>SWIPE_JOB_NOT_OPEN</code>	409	Job not open or company not approved
<code>SWIPE_UNDO_EXPIRED</code>	409	Outside undo window
<code>MATCH_FORBIDDEN</code>	403	Company does not own job / no right-swipe
<code>MATCH_NOT_FOUND</code>	404	Match id missing
<code>INVALID_SWIPE_PAYLOAD</code>	400	Zod validation

5. Locked Product Rules

- **One swipe per student per job** — enforced by DB unique (`student_id, company_id, job_id`); API returns 409 on replay.

- **Right ≠ match** — a student right-swipe alone does not create a match; company reciprocation (or an explicit Phase 7 match endpoint) does.
- **Feed hygiene** — after any swipe, that job never reappears on Discover for that student.
- **Save is independent** — Phase 6 saved_jobs remain bookmarks; swiping does not auto-save (unless product later chooses otherwise — out of Phase 7 default).
- **Match writes are server-owned** — clients never insert into matches via anon key; service_role through the module (same as Phase 3B RLS design).

6. Milestone Details

Milestone B1 — Swipes Module Scaffold

Goal: Create `src/modules/swipes/` (and thin matches routes or sibling module) with contracts, validation, and auth gates. Handlers may return 501 `NOT_IMPLEMENTED` until B2–B5. **Depends on:** Phase 2 auth, Phase 3B repos.

1. Add constants, types, Zod schemas, error codes, mapper stubs.
2. Mount router under `/api/v1` (swipes + matches paths).
3. Gate: student-only for POST `/swipes`; company+verified for inbound/match create.
4. Add `phase7.scaffold.test.ts`; script `npm run test:phase7`.

Exit checklist

#	Criterion	Done when
1	Module mounted	Routes reachable; 401 without token
2	Role gates	Company POST <code>/swipes</code> → 403
3	Contracts typed	DTOs match §4

Milestone B2 — Record Swipe + Feed Exclusion

Goal: Persist left/right swipes; Discover feed excludes swiped jobs for students.

1. Implement `createSwipe`: load job via `jobsRepository`; require status open + approved company; insert via `swipesRepository.create`.
2. Map unique violation → `SWIPE_CONFLICT`.
3. Update `jobsService.listFeed` for STUDENT viewers: `excludeJobIds = listJobIdsByStudent(viewer.id)`.
4. HTTP tests: swipe right → job absent from subsequent GET `/jobs`.

Exit checklist

#	Criterion	Done when
1	Swipe persists	Row in swipes after POST
2	Deck shrinks	Swiped job not in feed
3	Closed job blocked	<code>SWIPE_JOB_NOT_OPEN</code>

Milestone B3 — Undo Window (Optional)

Goal: Allow DELETE of a swipe within a short TTL (config, e.g. 30 seconds) so Discover can offer “Undo”. If skipped, document as deferred and keep UI without undo.

1. Add repository `deleteByStudentAndJob` (or soft-delete if preferred — default hard delete).
2. Service checks `swiped_at` + TTL; else `SWIPE_UNDO_EXPIRED`.
3. If match already created from that swipe, refuse undo (409) or cascade policy — lock: refuse.

Exit checklist

#	Criterion	Done when
1	Undo works	Within TTL, swipe gone and job may return to feed
2	Expired blocked	Outside TTL → 409

Milestone B4 — Company Interest + Match Creation

Goal: Verified companies list students who right-swiped their jobs and can create a match.

1. GET `/swipes/inbound` — filter `direction=right`, `company_id=caller`; join student public summary + job.

2. POST /matches — verify job ownership; require findStudentRightSwipe; matchesRepository.create or return existing (idempotent).
3. Optionally insert notifications (type=match) for both parties if notificationsRepository is ready.
4. Do not open conversations here unless already trivial — prefer Phase 8 for chat thread creation (or create empty conversation row if seed model requires it; document the choice).

Exit checklist

#	Criterion	Done when
1	Inbound list	Company sees right-swipes on own jobs
2	Match create	POST creates matches row once
3	Ownership	Other company's job → 403

Milestone B5 — Matches Read APIs

Goal: Students and companies list their matches with enough nested data for a Matches screen.

1. GET /matches/me — role-aware listByStudent / listByCompany.
2. Map to MatchDto with job title + counterparty name/logo.
3. No message history in this phase.

Exit checklist

#	Criterion	Done when
1	Student list	Returns matches for seed/demo accounts
2	Company list	Symmetric

Milestone B6 — Swagger, Hardening & Test Matrix

1. Document all \$4 paths in swagger.ts.
2. Add phase7.swagger + phase7.matrix tests; keep Express 5 query validation pattern (defineProperty, not Object.assign) from Phase 6 hotfix.
3. Ensure test:phase6 still green.

Exit checklist

#	Criterion	Done when
1	OpenAPI	Swipe/match paths at /api/docs
2	test:phase7	Full suite green

Milestone F1 — swipesApi & matchesApi

1. Add types + clients under frontend/src/lib/api/.
2. Map Phase 7 error codes in errorMessages.ts.

Exit checklist

#	Criterion	Done when
1	Clients export	Importable from lib/api
2	Types compile	tsc clean

Milestone F2 — Discover Live Swipe

1. Wire Discover Like/Nope (and drag end) to swipesApi.create.
2. On success, advance deck; on conflict, treat as already swiped and skip.
3. Refetch or locally filter so swiped jobs do not return after refresh.
4. Remove or rewrite // MOCK: Phase 7 comments on persistence.

Exit checklist

#	Criterion	Done when
1	Persist	Reload does not re-show swiped job
2	Errors	ErrorBanner on failure

Milestone F3 — Company Interest Inbox

1. New screen or section (e.g. under Applicants stub / new Interest route) listing inbound right-swipes.
2. Action: Match → POST /matches; update row state.
3. Link from Company Dashboard quick actions.

Exit checklist

#	Criterion	Done when
1	Inbox live	Shows student interest
2	Match action	Creates match visible on /matches/me

Milestone F4 — Matches Page Live

1. Replace prototype MatchesPage mock list with matchesApi.listMine.
2. EmptyState when none; click-through to job detail where useful.
3. Chat button may remain disabled with Phase 8 label.

Exit checklist

#	Criterion	Done when
1	List live	Seed/reciprocated matches appear
2	Mock removed	No hardcoded MATCHES array for data

Milestone F5 — Polish & Documentation

1. Update frontend/INTEGRATION.md live-vs-mock table (Discover swipe / Matches / Interest).
2. Write kupc-backend/documentation/PHASE_7_DOCUMENTATION.md as milestones ship.
3. Manual smoke: student swipe → company match → both see Matches.

Exit checklist

#	Criterion	Done when
1	Docs	INTEGRATION + PHASE_7_DOCUMENTATION
2	Smoke	End-to-end path works on seed accounts

7. Phase 7 Exit Checklist (Whole Phase)

#	Criterion	Owner
1	B1–B6 complete; npm run test:phase7 green	Backend
2	Swagger documents swipe + match endpoints	Backend
3	Student swipe persists; feed excludes that job after reload	Both
4	Duplicate swipe → 409 SWIPE_CONFLICT	Backend
5	Company inbound + reciprocate creates match	Both
6	GET /matches/me works for student and company	Both
7	Discover UI no longer local-only for swipe persistence	Frontend
8	Matches screen not using mock MATCHES seed array	Frontend
9	INTEGRATION.md + PHASE_7_DOCUMENTATION.md written	Both
10	test:phase6 still green; frontend tsc clean	Both

7.1 Manual smoke script

1. Login seed.student.001 → Discover → Like (right) an open job.
2. Refresh Discover → same job must not reappear.
3. Login seed.company (job owner) → Interest inbox → Match that student.
4. Both accounts → Matches → see the new match.
5. Student tries to swipe same job again → conflict handled gracefully.
6. Pending company must not access inbound/match mutating routes.

8. Phase 8 Preview — Chat & Conversations

After Phase 7, open conversation threads per match (conversations / messages already in Phase 3B), wire the Chat UI, and optionally emit match notifications. Discover/Matches remain the entry points; Phase 7 must not invent ad-hoc chat APIs.

End of KUPC Phase 7 Specification · Generated July 19, 2026 · Implement milestones in order; update PHASE_7_DOCUMENTATION.md as each milestone ships.